

Administrator Manual

This manual covers installation, licensing, multi-company setup, user and permission management, backups, and troubleshooting — everything beyond day-to-day use, which is covered in the User Manual.

1. Installation

See the Installation Guide for the full step-by-step procedure (system requirements, Setup.exe vs. Portable.exe, first-run activation, and verifying the install). Summary: run Setup.exe and follow the wizard (administrator rights are needed for this one-time step only), or run Portable.exe directly with no installation step at all. No Python installation or other runtime is required on the target machine — either option bundles everything the application needs.

1.1 Where application data lives

The application never writes inside its own install folder (Program Files is not writable by a standard user, and the goal is also to survive upgrades cleanly). All writable data — the database, logs, backups, uploaded files, and the license — lives under:

```
%LOCALAPPDATA%\AttendanceManagementSystem\  
├─ data\      (SQLite database)  
├─ logs\      (application logs, useful for troubleshooting)  
├─ backups\   (database backups)  
└─ uploads\   (employee photos, generated QR codes/barcodes, report exports)
```

This folder is created automatically on first launch and is never touched by installing, upgrading, or uninstalling the application — see "Upgrading" and "Uninstalling" below.

1.2 Upgrading

Running a newer Setup.exe over an existing installation performs an in-place upgrade: program files under the install directory are replaced, and %LOCALAPPDATA%\AttendanceManagementSystem\ (database, license, logs, backups) is left completely untouched. No manual data migration step is needed for a standard upgrade.

1.3 Uninstalling

Uninstall from Windows Settings → Apps, or via the Start Menu shortcut. The uninstaller removes only the installed program files. It then asks separately whether to also delete your data folder (%LOCALAPPDATA%\AttendanceManagementSystem\) — choose No to keep your database, license, and logs in place for a future reinstall, or Yes to fully remove all traces of the application and its data from the machine.

2. Licensing

The application requires an activated license before the login screen is shown. License keys are cryptographically signed (Ed25519) and bound to the specific machine they're activated on — a key generated for one computer will not activate on another.

2.1 Activation

On first launch (or whenever no valid license is present), the activation screen appears. Enter the license key you were provided and click Activate. On success, the application proceeds to the login screen; every subsequent launch skips this screen as long as the license remains valid.

2.2 License types

Type	Behavior
Trial	Time-limited; expires after its trial period.
Monthly / Yearly	Subscription-style; renewable, expires on the configured date.
Lifetime	Never expires.

2.3 Checking license status

The License Info screen (accessible from within the application once logged in — check the Settings area or your user menu) shows the current license type, expiry date (if any), and days remaining. Renew before expiry to avoid an interruption — contact your software vendor for a new key when a subscription is nearing its end.

2.4 Reactivation / transferring to a new machine

License keys are bound to the machine they were activated on. If you need to move the software to new hardware, contact your vendor for a reactivation — this application does not generate license keys itself; key generation is a vendor-side process kept separate from the shipped product for security (the signing key is never distributed with the application).

3. Multi-Company Setup

This is a true multi-tenant application: one installation, one database, any number of independent companies, each fully isolated from the others (no employee, department, user, device, or report from one company is ever visible to another). To onboard a new company, use the company management flow reachable from the login screen / initial setup — no code changes or per-company deployment is required.

Each company has its own:

- Employees, departments, branches, and shifts.
- Users, roles, and permissions.
- Devices and attendance records.
- Settings (time zone, date format, currency, backups).

3.1 Branches

Branches represent your company's physical locations/sites — separate from departments, which represent organizational structure. An employee or a device can optionally be assigned to a branch, letting you track where staff work and where a device is physically installed when a company operates from more than one location. Exactly one branch per company

can be marked as the "main branch" at a time; designating a new one automatically un-marks the previous one. Branch management is its own permission-gated screen (Branches), independent of Departments.

4. Users, Roles & Permissions

- Roles are per-company and fully customizable — there is no fixed global role list beyond what you define.
- Permissions are granted per role from a fixed catalog covering every module (dashboard, employees, departments, branches, attendance, devices, shifts, holidays, leave, reports, users/roles, settings, backups), each typically split into `.view` and `.manage` (or `.export`) grades — e.g. a role can be allowed to view attendance without being able to edit it, or view reports without being able to export them.
- Permissions are enforced at runtime, not just reflected in the menu. Every action — in the desktop app and in the optional REST API alike — checks the acting user's granted permission codes before touching the database. A user without a permission cannot perform the corresponding action even by some other route into the same feature; they simply don't see the corresponding screen or sidebar entry in the first place, since a screen is only ever built for permissions the logged-in user actually holds.
- Audit log: every sensitive operation (logins, record changes, backups, permission changes) is recorded with who performed it and when, for accountability and troubleshooting.

To create a new user: go to Users → Add, assign a company (implicit — you manage users within the company you're logged into), a username, an initial password, and a role. The user's password is hashed (bcrypt) before storage; there is no way to view a user's existing password, only to reset it.

4.1 Account lockout & sessions

Accounts lock temporarily after repeated failed login attempts, to resist password-guessing. Sessions expire automatically after a period of inactivity, requiring the user to log in again — this cannot be disabled from the UI and is a deliberate security default.

5. Devices

Supported protocols: ZKTeco (TCP/UDP push/pull protocol) and Hikvision (ISAPI over HTTP). For each device you configure its IP address and port; use "Test Connection" before saving to confirm reachability. Punch synchronization pulls new records from the device; pushing employee data enrolls them for recognition on the device's own hardware (biometric/card enrollment itself always happens on the device, not in this application).

Network requirements: the computer running this application must be able to reach each device's IP/port — same subnet, or routed and allowed through any firewalls in between. If a device intermittently drops, check for DHCP lease changes on the device (a static IP or DHCP reservation on the device is strongly recommended).

5.1 Automatic device synchronization

In addition to manually clicking "Sync," the application can pull new punches from every active device on a schedule, in the background, with no user action required. This is

controlled by environment variables (there is no in-app toggle, since it is a deployment-level setting):

Variable	Default	Meaning
DEVICE_AUTO_SYNC_ENABLED	true	Whether scheduled sync runs at all
DEVICE_AUTO_SYNC_INTERVAL_MINUTES	15	How often it runs

A failure syncing one device (e.g. temporarily unreachable) never blocks syncing the others — each device's result is independent, and failures are recorded in the application log rather than shown as a pop-up, since no one is necessarily watching the screen when a scheduled sync runs.

6. Backup & Restore

- Manual backup: from Settings → Backup, create a backup on demand. Backups are SQLite-WAL-compatible database snapshots stored under %LOCALAPPDATA%\AttendanceManagementSystem\backups\, encrypted at rest the same way sensitive database columns are (see 6.2 below) — a stolen backup file is not readable without the machine's own encryption key.
- Automatic scheduled backup: enabled by default (Settings → Preferences → "Auto Backup", or the BACKUP_AUTO_ENABLED / BACKUP_INTERVAL_HOURS environment variables) — the application checks periodically whether enough time has passed since the last backup and, if so, creates one and applies the retention policy (older backups beyond BACKUP_RETENTION_COUNT, default 14, are pruned automatically). You do not need to remember to back up manually once this is on.
- Restore: from the same screen, restore from a previously created backup file. Restoring replaces the current database — make a fresh backup of the current state first if you might need to go back to it.
- Recommended practice: back up before any major change (bulk employee import, department restructuring, a software upgrade) and keep periodic backups copied off-machine (network share, cloud storage) in case of hardware failure — the application's own backup feature protects against data mistakes, not against the machine itself being lost.

6.1 Data encryption

Certain sensitive columns — employee salary and device communication keys — are encrypted at rest in the database, not stored as plain text, using a per-installation encryption key generated automatically on first run and stored alongside the database (%LOCALAPPDATA%\AttendanceManagementSystem\data\field_encryption.key). This key is what both live encrypted columns and encrypted backups are protected with — losing it means an encrypted backup from this machine cannot be restored elsewhere, so back up that key file itself along with your regular database backups if you rely on off-machine backup restoration. It never needs to be entered or managed manually in normal operation.

7. Optional REST API

A separate, optional process (not part of the desktop application, and not started automatically) exposes the same company data over HTTP for external integrations — payroll systems, custom dashboards, mobile companion apps. It is disabled by default.

To enable it: set `API_ENABLED=true` (and optionally `API_HOST/ API_PORT`, default `127.0.0.1:8000`) in the environment, then run `python run_api.py` from the application's installation — this is a separate, deliberate step; installing or running the desktop application alone never opens a network port.

Authentication works the same way as the desktop login (username, password, and company), returning a bearer token used on every subsequent request; the same roles and permissions granted through Users/Roles govern what each API caller can do, with no separate permission system to maintain. Set a strong, unique `APP_SECRET_KEY` before enabling the API in any environment reachable by anyone other than trusted administrators — this key signs every issued token, and the built-in default is a placeholder meant to be overridden, not a real secret.

8. Troubleshooting

Symptom	Likely cause / fix
App won't start, no window appears	Check <code>%LOCALAPPDATA%\AttendanceManagementSystem\logs\</code> for the most recent log file and the error it recorded.
database connection error) at startup) "البيانات تعذر الاتصال بقاعدة"	The data folder or database file may be missing/corrupted, or in use by another process. Restore from a recent backup if needed.
Device shows disconnected	Confirm the device is powered on and reachable (ping its IP); confirm the IP/port configured in Devices still matches the device.
Scheduled device sync / backup never seems to run	Confirm the relevant <code>*_ENABLED</code> environment variable isn't set to <code>false</code> , and that the application process is actually kept running (it only runs while the desktop app is open — closing it stops the schedule until next launch).
Arabic text garbled in an exported PDF	Should not occur in a correctly built release — the app bundles its own Arabic-capable font and text-shaping libraries. If it does, the installation may be corrupted; reinstall.
Login screen shows no companies	No company has been set up yet, or all companies are marked inactive — check company setup / reactivate the company.
User locked out	Wait for the lockout window to pass, or have another administrator reset the account.
License shows expired	Contact your vendor for a renewed key and re-activate via the activation screen.
REST API returns 401 for a request you expect to succeed	Confirm the bearer token hasn't expired (<code>API_TOKEN_EXPIRES_MINUTES</code>) and that you're sending <code>Authorization: Bearer <token></code> exactly as returned by <code>/api/auth/login</code> .

Symptom	Likely cause / fix
REST API returns 403 for a request you expect to succeed	The logged-in user's role doesn't have the required permission — grant it via Users → Roles, the same as you would for the desktop app.

For issues not covered here, collect the relevant log file(s) from %LOCALAPPDATA%\AttendanceManagementSystem\logs\ before contacting support — they contain the detail needed to diagnose most problems quickly.